TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Validated high-power lithium titanate oxide (LTO) battery modelling, safety marker detection and BMS optimisation for fast-charging applications			
Korea's 12 National Strategic Technology Fields	Semiconductor and Display ( ) Secondary Battery (√) Advanced Mobility (√) Next generation nuclear power ( ) Advanced biotechnology ( ) Aerospace and marine technology ( ), Hydrogen ( ) Cybersecurity ( ) Artificial intelligence (AI) ( ) Next generation communications ( ) Advanced robotics and manufacturing ( ) Quantum technology ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Instrumentation	Safety monitoring/diagnosis measurement and control instruments
Concept of Proposed Technology	The proposed technology focuses on the joint development and validation of a high-power lithium titanate oxide (LTO) battery module integrated with an intelligent battery management system (BMS) optimized for fast charging, safety monitoring, and early-stage fault detection. The core concept is to treat the battery module and BMS as a co-designed cyber-physical			

	system, rather than as independently optimized components, enabling reliable operation under aggressive charging conditions [2][3]. At the hardware level, the technology targets LTO-based cells and modules due to their demonstrated advantages in fast-charge capability, thermal robustness, and cycle life under high-power operation. However, experimental evidence shows that sustained fast charging at high C-rates requires coordinated control of thermal behaviour, voltage limits, and ageing mechanisms, which cannot be achieved through hardware design alone [5][6][7]. The proposed BMS therefore integrates electro-thermal and ageing-aware battery models to dynamically adapt charging strategies and protection limits in real time [2][5]. At the software and algorithmic level, the BMS implements advanced state estimation (SOC, SOH), abnormal behaviour detection, and safety marker logging. These functions extend beyond conventional rule-based protection by combining model-based estimation with data-driven diagnostics, enabling early identification of deviations associated with degradation, thermal runaway precursors, or sensor inconsistencies [2][4]. Explainable diagnostics and structured data logging are explicitly included to support traceability, validation, and future certification processes [1][4]. Validation is a central element of the concept. The integrated module–BMS system is verified through laboratory characterisation, accelerated ageing tests, safety-oriented experiments, and hardware-in-the-loop (HIL) validation, ensuring that performance claims are supported by reproducible data rather than simulation alone [1][2]. This validation-driven approach enables the delivery of a scalable and application-agnostic reference architecture for fast-charging LTO battery systems suitable for high-power use cases where safety and reliability are critical.
Rationale for Joint R&D Project	Fast-charging battery systems are a key enabler for electrification in mobility, logistics, and industrial applications, but they introduce significant challenges related to thermal stress, accelerated degradation, and safety assurance. Although LTO chemistry has shown strong intrinsic tolerance to high C-rate charging, published studies and applied research consistently indicate that reliable fast-charging performance depends on tightly integrated battery design, BMS algorithms, and experimental validation [5][6][7]. A joint R&D project is therefore required to bridge the gap between laboratory feasibility and deployable, validated systems. Innovative SMEs typically possess strong capabilities in battery hardware development and application-specific system integration, but often lack access to long-duration ageing infrastructure, advanced battery modelling expertise, and systematic safety validation methodologies [1][3]. These gaps represent a major barrier to de-risking fast-charging technologies for commercial deployment.

	The proposed collaboration combines SME-driven system innovation with applied research institute expertise in battery characterisation, electro-thermal modelling, safety diagnostics, and HIL-based verification. Joint development enables iterative refinement of charging strategies and diagnostic algorithms based on experimental evidence, accelerating progression from laboratory-scale validation (TRL 4) to system-level reliability demonstration (TRL 7) [1][2]. In addition to technical risk reduction, the joint R&D approach supports knowledge transfer and the creation of reusable methods for battery safety marker detection, remaining useful life estimation, and ageing-aware charging control, building on prior applied research activities in advanced BMS development [4][5][6]. The resulting validated prototypes, test protocols, and model-based control frameworks provide a concrete foundation for follow-on industrialisation, certification planning, and market entry by participating SMEs, while strengthening international collaboration in the secondary battery domain [1][9].
Key Words	1. Lithium titanate oxide (LTO) 2. fast charging 3. battery management system (BMS) 4. safety diagnostics 5. state estimation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	Proposed objectives: • Define target requirements and baseline metrics for LTO modules and BMS. • Develop BMS algorithms for fast charging, SOC/SOH estimation, protection and diagnostics. • Build and validate electro-thermal and ageing models to optimise charging strategies. • Prototype and validate integrated module + BMS in lab tests (electrical, thermal, ageing, safety) and HIL. • Deliver a verified prototype data package to support deployment and certification planning.			
R&D Target Specifications	Target specifications (proposed and to be finalised with the Korean SME partner): • Demonstrate repeated fast charging of an LTO module at 4C or higher, with controlled temperatures. • Benchmark target: charge to 80% SOC within 15 minutes (application dependent). • Implement cell monitoring, balancing, and protection functions suitable for high-power operation. • Implement abnormal behaviour detection and safety marker logging with actionable thresholds.			
R&D Plan	Proposed 36-month work plan (timeline and milestones to be aligned with the selected TIPA track): **WP1 – Requirements and Data Preparation (Month 0–6)** • Define functional, safety, and KPI requirements for early battery fire detection • Establish access to BMS and charging data and perform preprocessing • Prepare representative normal-operation datasets and baseline features **WP2 – Interpretable Anomaly Detection Model (Month 6–15)** • Develop and train an Interpretable AutoEncoder using normal-operation data • Implement reconstruction-error–based anomaly scoring • Enable explainability through signal-level and temporal attribution **WP3 – Abnormal Scenario Validation and Model Refinement (Month 12–24)** • Generate abnormal and degradation scenarios using controlled battery tests • Integrate experimental and field data to improve robustness • Refine adaptive thresholds and risk scoring across operating conditions **WP4 – Real-Time System and Visualization (Month 18–30)** • Implement real-time data ingestion and anomaly detection pipeline • Integrate early-warning logic in edge/cloud architecture • Develop monitoring dashboard with interpretable alerts			

	WP5 – Pilot Demonstration and Exploitation (Month 24–36) • Validate system performance in pilot operational environments • Verify KPIs and reliability towards TRL 7 • Assess commercialization pathway for battery safety management SaaS

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) proposed role and relevant technical capabilities: • Provide access to the TNO Battery Lab for accelerated ageing tests, battery characterisation, safety research and prototyping of modules and packs. [1] • Develop and validate battery models and BMS algorithms including SoX estimation, balancing algorithms, thermal state estimation/prediction and ageing-aware charging control. [2][3] • Use model-in-the-loop and (hardware-in-the-loop) verification, then embed algorithms in pack prototypes for lab validation under representative charging conditions. [1][2] • Apply TNO methods for Remaining Useful Life prediction and early abnormality detection (safety markers), drawing on experience from NEXTBMS-related activities. [1][4]
Expected R&R of Korean SME Partner	Korean SME partner - proposed role: • Provide LTO cell and module designs, manufacturing capability, and application requirements. • Provide or co-develop BMS hardware and embedded implementation as appropriate. • Build prototypes and support pilot production readiness. • Lead commercialisation planning for target markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes (proposed): - Prototype LTO module(s) with integrated BMS and documented fast-charge operating envelope. - Validated test protocols and data sets (electrical, thermal, ageing and safety). - Model and algorithm package for SOC/SOH estimation and diagnostics. - Joint IP and publications where appropriate, plus an exploitation pathway. Expected impact (proposed): - Accelerated development of safe fast-charging battery module and intelligent management technologies. - Strengthened Korea-EU collaboration in secondary battery modules and systems.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization